

IoT Challenge #3 – Part 2 (Exercise)

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Scenario and Parameters

The LoRaWAN network operates in Europe at carrier frequency 868 MHz with bandwidth 125 kHz. It has one gateway and:

$$N = 40$$

sensor nodes. Each node transmits packets with payload size 20 bytes according to a Poisson process with intensity:

$$\lambda = 2 \text{ packets/minute} = \frac{2}{60} \text{ packets/second} = \frac{1}{30} \text{ packets/second}.$$

All nodes are assumed to use the same spreading factor. Following the simplified ALOHA model used in the course material, the packet success rate is:

$$P_{\text{success}} = \exp(-2N\lambda T_{\text{air}}),$$

where T_{air} is the packet airtime in seconds.

EQ1: Largest SF with at Least 75% Packet Success Rate

The target condition is:

$$\exp(-2N\lambda T_{\text{air}}) \geq 0.75.$$

Solving it for T_{air} gives:

$$\begin{aligned} T_{\text{air}} &\leq \frac{-\ln(0.75)}{2N\lambda} \\ &= \frac{-\ln(0.75)}{2 \cdot 40 \cdot \frac{1}{30}} \\ &= 0.10788 \text{ s} \\ &= 107.88 \text{ ms.} \end{aligned}$$

Therefore, the largest acceptable spreading factor is the largest SF whose airtime is not greater than about 107.88 ms.

Airtime Calculation

The airtimes below are computed using the TTN LoRaWAN airtime calculator (<https://www.thethingsnetwork.org/airtime-calculator>) with the parameters specified by the assignment:

- input bytes: 20;
- region: Europe 863–870 MHz;
- bandwidth: 125 kHz;
- spreading factor varied from SF7 to SF12.

The TTN calculator is a LoRaWAN airtime calculator: the 20 input bytes are the application payload bytes, and the calculator includes the LoRaWAN overhead in the time-on-air result. The carrier frequency does not affect airtime.

SF	TTN time on air [ms]	ALOHA success rate
7	71.9	82.54%
8	133.6	70.02%
9	246.8	51.78%
10	452.6	29.91%
11	987.1	7.19%
12	1810.4	0.80%

Table 1: TTN airtime and ALOHA success rate for 20 input bytes at BW 125 kHz.

SF7 is below the maximum allowed airtime:

$$T_{\text{air,SF7}} = 71.9 \text{ ms} < 107.88 \text{ ms},$$

and its success rate is:

$$P_{\text{success,SF7}} = \exp\left(-2 \cdot 40 \cdot \frac{1}{30} \cdot 0.0719\right) \approx 0.8254.$$

SF8 is already above the airtime threshold when using the TTN LoRaWAN airtime calculator:

$$T_{\text{air,SF8}} = 133.6 \text{ ms} > 107.88 \text{ ms},$$

and its success rate drops to about 70.02%.

Answer EQ1: Using the TTN LoRaWAN airtime calculator with 20 input bytes, the largest spreading factor that guarantees at least 75% packet success rate is SF7.

EQ2: Field Deployment with Non-Uniform Performance

The measured packet success rate is significantly lower than expected, and the performance is not uniform across nodes: some nodes work relatively well, while others have very poor delivery rates.

The most appropriate action is:

2. Move the nodes closer to the gateway.

The non-uniformity across nodes suggests a coverage or link-budget problem. Nodes that are farther from the gateway, or in worse propagation conditions, may have received power close to or below the receiver sensitivity, causing poor delivery even if the collision model predicted a better average success rate.

Decreasing the number of nodes would reduce the offered traffic and, therefore, the collision probability, but it would mainly address a global congestion problem. It does not directly explain why only some nodes perform very poorly.

Increasing the spreading factor can improve sensitivity and range, but it also increases packet airtime. Longer airtime increases the probability of overlap and collisions in the ALOHA model. Therefore, increasing SF is not the best first action when the problem is non-uniform and likely related to node placement or coverage.

Answer EQ2: Move the nodes closer to the gateway, because the observed non-uniform delivery rate points to coverage/link-budget issues rather than only network-wide congestion.